INTERNSHIP PROJECT REPORT

Task 1: Student Management System (Python OOP & Persistence)

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Application ID:	KTS020260716223
Internship Track:	Python Development Internship
Duration:	30 Days (20 July 2026 – 19 August 2026)
Supervisor / HR:	Prince Asthana (HR Executive)

1. Executive Summary

The Student Management System is a modular, production-ready Python application engineered to streamline academic record keeping. The solution adheres strictly to Object-Oriented Programming (OOP) principles, clean architecture patterns, defensive validation, robust exception handling, dual persistence layers (JSON & CSV), and provides an interactive terminal user interface.

2. Key Architectural Features & Capabilities

- **Object-Oriented Design (OOP):** Domain models (Student, Course) featuring properties, private member encapsulation, and custom serialization methods.
- **Complete CRUD Operations:** Create, Read, Update, and Delete operations for student records with rollback safety.
- **GPA & Academic Tracking:** Dynamic weighted GPA computation based on course credit hours and numerical grade points.
- **Search & Multi-criteria Filtering:** Instant substring search across roll numbers, names, departments, and emails, with GPA and status filters.
- **Defensive Validation & Custom Exceptions:** Regex-based email and phone verification, non-empty checks, and custom error classes (DuplicateRollNumberError, ValidationError).
- **Dual Persistence Engine:** Atomic JSON file operations with temporary swap protection and CSV import/export capabilities.
- **Automated Test Suite:** Comprehensive Pytest unit tests achieving high coverage of domain and service layer logic.

3. Module Structure

Layer / Module	File Path	Responsibility
Domain Models	src/models/student.py	Encapsulates Student state, GPA computation, course enrollment.
Course Entity	src/models/course.py	Course definitions, credit weighting, and grade points.
Storage Layer	src/storage/json_storage.py	Atomic JSON persistence with crash recovery mechanism.
Export / Import	src/storage/csv_storage.py	Bulk CSV data import and export operations.
Service Layer	src/services/student_service.py	Core business logic, search, filter, and analytics.
CLI Interface	src/ui/cli.py	Interactive menu-driven terminal interface with styled tables.
Test Suite	tests/test_student_service.py	Automated validation for CRUD, duplicates, and edge cases.

4. Verification & Testing

The project was verified using Pytest. All unit tests covering student registration, duplicate prevention, data format validation, GPA calculation, search/filter algorithms, and storage persistence executed successfully.

5. Learning Outcomes & Conclusion

During this assignment, industry best practices in Python development were implemented, including modular project architecture, SOLID principles, type annotations, unit testing, and automated documentation. The application is fully extensible for future database (SQL/NoSQL) and REST API integrations.